

APM 1110 - FA 2

Cainglet, Manalaysay

<https://github.com/hamiangg/APM-1110---FAs-SAs>

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Amy and Jane are gambling together. A fair coin is tossed repeatedly. Each time a head comes up, Amy wins two euro from Jane, and each time a tail comes up, Amy loses two euro to Jane.

Simulates coin toss 100 times (with replacement)

```
tosses <- sample(c("H", "T"), size = 100, replace = TRUE)
```

Amy wins two euros for head but loses two euros for tail

```
win_lose <- ifelse(tosses == "H", 2, -2)
```

Determine Amy's current balance after each toss

```
balance <- cumsum(win_lose)
```

4a.

The number of times that Amy is ahead in these 100 tosses

```
ahead <- sum(balance > 0)
```

```
ahead
```

```
## [1] 79
```

4b.

How much Amy has won or lost after 100 tosses

```
total <- balance[100]
if (total > 0) {
  paste("Amy won €", total)
} else if (total < 0) {
  paste("Amy lost €", abs(total))
} else {
  "Amy broke even (€0)"
}
```

```
## [1] "Amy won € 20"
```

```
total
```

```
## [1] 20
```

Plot of Amy's Winning

```
plot(balance, type = "o", col = "hotpink",
      main = "Amy's Winnings per Coin Toss",
      xlab = "Toss Number", ylab = "Winnings (€)")
abline(h = 0, lty = 2)
```

